

Q1 Reverse a list

1 mark

Write a C program that takes an array as input and returns it reversed.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5     printf("Enter the number of elements: ");
6     scanf("%d", &n);
7
8     int arr[n];
9     printf("Enter the elements: ");
10    for (i = 0; i < n; i++) {
11        scanf("%d", &arr[i]);
12    }
13
14    printf("Reversed array: ");
15    for (i = n - 1; i >= 0; i--) {
16        printf("%d ", arr[i]);
17    }
18    printf("\n");
19
20    return 0;
21 }
```

Input

```
5
1 2 3 4 5
```

Output

```
Enter the number of elements: Enter
the elements: Reversed array: 5 4 3
2 1
```

Q2 Reverse a string

1 mark

Write a C program for palindrom

Your Code

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int isPalindrome(const char *s) {
5     int i = 0, j = (int)strlen(s) - 1;
6     while (i < j) {
7         if (s[i] != s[j]) return 0;
8         i++;
9         j--;
10    }
11    return 1;
12 }
13
14 int main() {
15     char str[1000];
16
17     printf("Enter a string: ");
18     scanf("%999s", str);
19
20     if (isPalindrome(str))
21         printf("Palindrome\n");
22     else
23         printf("Not Palindrome\n");
24
25     return 0;
26 }
```

Input

```
racecar
```

Output

```
Enter a string: Palindrome
```

Submitted

Q3 Searching

1 mark

Write a C program for Searching an Element in a Sorted Array

Your Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int n, int target) {
4     int low = 0, high = n - 1;
5
6     while (low <= high) {
7         int mid = low + (high - low) / 2;
8
9         if (arr[mid] == target)
10             return mid;
11         else if (arr[mid] < target)
12             low = mid + 1;
13         else
14             high = mid - 1;
15     }
16     return -1;
17 }
18
19 int main() {
20     int n, target;
21
22     printf("Enter number of elements: ");
23     scanf("%d", &n);
24
25     int arr[n];
26     printf("Enter sorted array elements: ");
27     for (int i = 0; i < n; i++) {
28         scanf("%d", &arr[i]);
29     }
30
31     printf("Enter element to search: ");
32     scanf("%d", &target);
33
34     int index = binarySearch(arr, n, target);
35
36     if (index != -1)
37         printf("Element %d found at index %d.\n", target, index);
38     else
39         printf("Element %d not found in the array.\n", target);
40
41     return 0;
42 }
```

Input

```
5
2 4 6 8 10
8
```

Output



Q4 Searching

Write a C program for Count the Occurrences of an Element

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, x, count = 0;
5
6     // Input size of array
7     scanf("%d", &n);
8
9     int arr[n];
10
11    // Input array elements
12    for (int i = 0; i < n; i++) {
13        scanf("%d", &arr[i]);
14    }
15
16    // Input element to search
17    scanf("%d", &x);
18
19    // Count occurrences
20    for (int i = 0; i < n; i++) {
21        if (arr[i] == x) {
22            count++;
23        }
24    }
25
26    // Print result
27    printf("%d", count);
28
29    return 0;
30 }
```

Write a program for binary search

Your Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int size, int target) {
4     int low = 0, high = size - 1;
5
6     while (low <= high) {
7         int mid = low + (high - low) / 2;
8
9         // Check if target is at mid
10        if (arr[mid] == target)
11            return mid;
12
13        // If target is greater, ignore left half
14        else if (arr[mid] < target)
15            low = mid + 1;
16
17        // If target is smaller, ignore right half
18        else
19            high = mid - 1;
20    }
21
22    // Target not found
23    return -1;
24 }
25
26 int main() {
27     int n, target;
28
29     printf("Enter the number of elements: ");
30     scanf("%d", &n);
31
32     int arr[n];
33
34     printf("Enter %d sorted elements:\n", n);
35     for (int i = 0; i < n; i++) {
36         scanf("%d", &arr[i]);
37     }
38
39     printf("Enter the element to search: ");
40     scanf("%d", &target);
41
42     int result = binarySearch(arr, n, target);
43
44     if (result != -1)
45         printf("Element %d found at index %d\n", target, result);
46     else
47         printf("Element %d not found in the array\n", target);
48
49     return 0;
50 }
```

Input

```
5
1 3 5 7 9
5
```

Output

```
Enter the number of elements: Enter
5 sorted elements:
Enter the element to search:
Element 5 found at index 2
```

Q6 Merging

1 mark

Write a C program for Merge Two Arrays

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n1, n2;
5
6     printf("Enter size of first array: ");
7     scanf("%d", &n1);
8     printf("Enter size of second array: ");
9     scanf("%d", &n2);
10
11     int a[n1], b[n2], c[n1 + n2];
12
13     printf("Enter elements of first sorted array:\n");
14     for (int i = 0; i < n1; i++) {
15         scanf("%d", &a[i]);
16     }
17
18     // Input second array
19     printf("Enter elements of second sorted array:\n");
20     for (int i = 0; i < n2; i++) {
21         scanf("%d", &b[i]);
22     }
23
24     // Merge
25     int i = 0, j = 0, k = 0;
26     while (i < n1 && j < n2) {
27         if (a[i] <= b[j]) {
28             c[k++] = a[i++];
29         } else {
30             c[k++] = b[j++];
31         }
32     }
33
34     // Remaining elements
35     while (i < n1) c[k++] = a[i++];
36     while (j < n2) c[k++] = b[j++];
37
38     // Output
39     printf("Merged array: ");
40     for (int t = 0; t < n1 + n2; t++) {
41         printf("%d ", c[t]);
42     }
43     printf("\n");
44
45     return 0;
46 }
```

Input

```
4
1 3 5 7 9
2 4 6 8
```

Output

```
Enter size of first array:
Enter size of second array:
Enter elements of first sorted
array:
Enter elements of second sorted
array:
```

Submitted

Q7 Largest and Smallest element

1 mark

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5
6     printf("Enter the number of elements: ");
7     scanf("%d", &n);
8
9     int arr[n];
10
11    printf("Enter %d elements:\n", n);
12    for (i = 0; i < n; i++) {
13        scanf("%d", &arr[i]);
14    }
15
16    int largest = arr[0];
17    int smallest = arr[0];
18
19    for (i = 1; i < n; i++) {
20        if (arr[i] > largest) {
21            largest = arr[i];
22        }
23        if (arr[i] < smallest) {
24            smallest = arr[i];
25        }
26    }
27
28    printf("Largest element: %d\n", largest);
29    printf("Smallest element: %d\n", smallest);
30
31    return 0;
32 }
```

Submitted

Input

```
5
10 3 25 7 2
```

Output

```
Enter the number of elements: Enter
5 elements:
Largest element: 25
Smallest element: 2
```

Q8 Second Largest Element

1 mark

Write a C program for Find the Second Largest Element

Your Code

```
1 #include <stdio.h>
2 #include <limits.h>
3
4 int main() {
5     int n;
6     if (scanf("%d", &n) != 1 || n < 2) {
7         printf("Second largest element not found");
8         return 0;
9     }
10
11     int firstMax = INT_MIN;
12     int secondMax = INT_MIN;
13
14     for (int i = 0; i < n; i++) {
15         int x;
16         scanf("%d", &x);
17
18         // Update firstMax and secondMax for distinct values
19         if (x > firstMax) {
20             if (x != firstMax) { // always true here, but keep
21                 secondMax = firstMax;
22                 firstMax = x;
23             }
24         } else if (x < firstMax && x > secondMax) {
25             secondMax = x;
26         }
27     }
28
29     if (secondMax == INT_MIN) {
30         printf("Second largest element not found");
31     } else {
32         printf("%d", secondMax);
33     }
34
35     return 0;
36 }
```

Input

25 50

Output

Q9 Occurrence of an element

Write a C program for Find the First and Last Occurrence of an Element

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, x;
5     int arr[100];
6     int first = -1, last = -1;
7
8     printf("Enter number of elements: ");
9     scanf("%d", &n);
10
11    printf("Enter array elements:\n");
12    for (i = 0; i < n; i++) {
13        scanf("%d", &arr[i]);
14    }
15
16    printf("Enter element to search: ");
17    scanf("%d", &x);
18
19    for (i = 0; i < n; i++) {
20        if (arr[i] == x) {
21            if (first == -1) first
```

Submitted

Q10 Missing Element

1 mark

Write a C program for Find Missing Element in an Array

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     printf("Enter size of array (n): ");
6     scanf("%d", &n);
7
8     // n elements given, but missing one from 1..n+1 (typical case)
9     int arr[n];
10
11    printf("Enter %d elements:\n", n);
12    for (int i = 0; i < n; i++) {
13        scanf("%d", &arr[i]);
14    }
15
16    // Using sum formula:
17    // Missing = (1 + 2 + ... + (n+1)) - (sum of array elements)
18    int expected_sum = (n + 1) * (n + 2) / 2; // sum from 1 to n+1
19    int actual_sum = 0;
20
21    for (int i = 0; i < n; i++) {
22        actual_sum += arr[i];
23    }
24
25    int missing = expected_sum - actual_sum;
26    printf("Missing element = %d\n", missing);
27
28    return 0;
29 }
```

Input

2

Output

Enter size of array (n): Enter 2
elements:
Missing element = 5